

1. For a discrete statistical model and an observed sample $\underline{x}$, suppose $L_{\underline{x}}(\theta_1) = 2L_{\underline{x}}(\theta_2) > 0$. Which conclusion follows?

A. The parameter is twice as likely to equal θ_1 as θ_2 .

☒ B. The observed sample has twice the probability under θ_1 as under θ_2 .

C. θ_2 cannot be the true parameter.

D. The probability that $\theta = \theta_1$ is $2/3$.

2. Suppose $T(\underline{X})$ is minimal sufficient for θ , and $S(\underline{X})$ is another sufficient statistic. Which statement is always true?

A. S is a function of T .

B. The distribution of T does not depend on θ .

☒ C. T is a function of S .

D. Every function of T is sufficient for θ .

3. Let X_1, X_2 be independent Bernoulli(p) random variables, where $0 < p < 1$, and let $T = \max(X_1, X_2)$. What is $\mathbb{P}_p((X_1, X_2) = (1, 1) \mid T = 1)$, and what does it imply about sufficiency?

A. $1/3$; T is sufficient for p .

B. p^2 ; T is not sufficient for p .

☒ C. $p/(2 - p)$; T is not sufficient for p .

D. 1 ; T is sufficient for p .